A certified partial census of bistability in three-species open bimolecular mass-action networks with at most three internal reactions

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AI-assistance disclosure and audit status. This preprint, its search and checking code and its claim ledger were produced by AI agents (Claude, Anthropic) under the author's direction; the author is responsible for the content. **The census was independently audited once by a different AI agent instance (`data/AUDIT.md`, ACCEPT_WITH_CORRECTIONS), not by a human.** The audit's corrections are applied (Section 7). **No human audit exists.** The preregistration hash is self-attested and has no independent timestamp.

Abstract

We enumerate all mass-action networks on three species X, Y, Z built from the nine complexes of molecularity at most two, with at most three internal reactions and, for every species, an inflow and a linear removal: 62,269 labelled networks, 10,523 up to species relabelling. For each we look for a certificate. 754 **networks carry an exact rational bistability witness** (positive rational parameters and two distinct positive rational equilibria, both exactly Routh–Hurwitz stable), independently re-verified by an auditing AI agent. 5,071 **carry a certified exclusion** (at most one positive equilibrium for all parameters, by a determinant-sign criterion of Craciun–Feinberg type, proved for our quadratic fields by a midpoint identity). 807 more have two exact equilibria but no stable pair was found (*not* independently verified), and 3,891 (37.0%) **are UNDECIDED**: the census is *partial*, and an undecided network is search evidence only, never an absence. Among connected-support networks, 727 are certified bistable, 175 of them minimal under edge deletion; of these 79 contain a species that is never a reactant and are effectively two-species cascades, and 96 are irreducible by our screen. Exactly nine networks with two internal reactions are bistable; they are listed with one exact witness each. The census reproduces Peng's two-species classification (arXiv:2610.32739, 27 of 681), which is Peng's result, not ours. A consistency check against the inheritance theorem of Joshi–Shiu shows that at least 57 further three-reaction networks are bistable although not certified here. We make no claim of new mechanisms and no biological claim; novelty relative to the cited literature is "to our knowledge" only (Section 8). Producer and auditor are AI agents; no human audit.

Markers. [L]: a cited source, taken as stated. [C]: executed code with an argument written down but not machine-checked (no Lean here). [X]: search output, a recorded absence, or a list of what was not done, supporting nothing by itself. There is no Tier A claim in this paper. Identifiers `GE2-·-nnnn` refer to `ledger/ledger.json`.

1 The class and its equations

Fix species $X_1, X_2, X_3 = X, Y, Z$. The nine *complexes* are $X, Y, Z, 2X, 2Y, 2Z, X{+}Y, X{+}Z, Y{+}Z$ (nonzero, molecularity ≤ 2). An *internal reaction* is a directed edge $\alpha \to \beta$ between two distinct complexes (72 edges). A *network* is a set of $m \leq 3$ distinct internal edges together with, for every species S, the inflow $0 \to S$ (rate a_S) and the removal $S \to 0$ (rate ρ_S). With mass action, writing $x = (x_1, x_2, x_3)$,

$$\dot{x}_i = f_i(x) = a_i - \rho_i x_i + \sum_e k_e \, (\beta_e - \alpha_e)_i \, x^{\alpha_e}, \tag{1}$$

where all parameters a_i, ρ_i, k_e are positive reals (no normalisation). A positive equilibrium is a zero of f in the open orthant; f is a polynomial of degree at most two. **Definition.** A network is *bistable* if some positive parameter vector gives two distinct positive equilibria that are both linearly stable (all Jacobian eigenvalues with negative real part). Stable equilibria are nondegenerate. Because every species has inflow and removal, the stoichiometric subspace is $\mathbb{R}^3$ and every equilibrium is in the single compatibility class.

Buckets. Bucket A: at most two species occur in the internal reactions (a species occurring only in flows decouples; these networks reduce to the two-species class of Peng [1]). Bucket B: all three species occur but the species-interaction graph (species joined if they occur in the same reaction) is disconnected. Bucket C: connected support on three species. "Connected support" does not mean "genuinely three-species": Section 5.1 shows that 79 of the 175 minimal bistable networks in bucket C contain a never-reactant species.

Two-species class (control). $\{X, Y, 2X, X{+}Y, 2Y\}$, 20 edges, $m \leq 3$: 1351 labelled networks, 681 orbits under S_2, exactly the class of Peng [1].

2 Enumeration and the Burnside count

`GE2-C-0001` [C]. Labelled networks with $m = 0, 1, 2, 3$ internal edges number $1 + 72 + \binom{72}{2} + \binom{72}{3} = 1 + 72 + 2556 + 59640 = 62269$. The symmetric group S_3 acts on species; the canonical form of a network is the lexicographic minimum, over the six permutations, of its sorted edge-index tuple. Orbits have sizes $1, 2, 3, 6$ ($7, 9, 266, 10241$ orbits) and the sizes sum to 62269; the orbit count is 10523, and Burnside's lemma gives 10523 independently. The auditor wrote its own enumerator (`data/code_auditor/enum.py`, no producer code) and obtained 62269, 10523 and, for the two-species class, 1351 and 681. The census has one row per orbit (`data/census.jsonl`). Bucket sizes: A 681, B 220, C 9622 orbits.

3 Exclusion: a determinant-sign criterion

The criterion belongs to the Jacobian/injectivity tradition for fully open networks (Craciun–Feinberg [5, 6]; Joshi–Shiu [7]; `GE2-L-0003` [L]). We do not claim it as new and did not read those papers in this session; we prove the version we use, which is short because f is quadratic.

Proposition 1. *Let f be as in* (1) *and let $J(x) = Df(x)$. If $\det(-J)$, as a polynomial in the variables (x, ρ, k), has only nonnegative coefficients and the coefficient of $\rho_1\rho_2\rho_3$ is* 1*, then for every positive parameter vector f has at most one zero in the open positive orthant.*

Proof. Let x, y be positive zeros. Since f has degree ≤ 2, Df is affine, so $\int_0^1 Df(y+t(x-y))\,dt = J(\frac{x+y}{2})$ and

$$0 = f(x) - f(y) = J\big(\tfrac{x+y}{2}\big)\,(x-y).$$

The midpoint $m = \frac{x+y}{2}$ is positive. By hypothesis $\det(-J(m))$ is a sum of products of positive quantities, with the term $\rho_1\rho_2\rho_3 > 0$ present, so $\det J(m) \neq 0$ and $x = y$. □

Here $-J = \operatorname{diag}(\rho) - \sum_e k_e(\beta_e - \alpha_e)\nabla(x^{\alpha_e})^\top$, and the coefficients of $\det(-J)$ are computed symbolically for each network (sympy). A network passing the test is labelled MONOSTATIONARY_JAC (5071 orbits, 28,158 + 1444 + 346 = 29,948 labelled; `GE2-C-0003` [C]). A second, sign-pattern rule (monomial-difference signs row by row) was implemented and *excluded nothing beyond* the determinant test. The auditor judged the argument sound, re-derived 60/60 random excluded rows (all coefficients nonnegative) and 40/40 random non-excluded rows (a negative coefficient), and a numerical search on 30 excluded rows (40 parameter draws, 30 starts each) found no second positive root. The remaining excluded rows were computed by the producer only and sampled by the auditor, not re-derived one by one.

4 Witness search and exact certification

Search (existence only). Two equilibria $p \neq q$ force a linear system in (ρ, k): $\rho_i(p_i - q_i) = \sum_e k_e(\beta_e - \alpha_e)_i(p^{\alpha_e} - q^{\alpha_e})$, and then $a_i = \rho_i p_i - H_i(p)$ is linear in (ρ, k). For each non-excluded network we sample pairs (p, q) log-uniformly in $[10^{-2}, 10^2]^3$ (half of the pairs with $q = p\exp N(0, 0.5)$ coordinatewise), take the kernel of that system by SVD, draw random kernel combinations, accept those with $\rho, k, a > 0$ and floating Routh–Hurwitz at p and q, rationalise p, q (denominator $\le 10^4$), solve the kernel exactly (sympy) and take a rational combination near the float vector. Budget: 2×10^5 pairs times 4 combinations per network, fixed seed per network.

Exact certificate (BISTABLE_CERT, `GE2-C-0002` [C]). Each witness file (`data/witnesses/G*.json`) holds rational a, ρ, k and two distinct positive rational equilibria p, q. The checker substitutes in exact fractions, verifies $f(p) = f(q) = 0$, and applies Routh–Hurwitz exactly at both points (3×3: $c_1 > 0$, $c_3 > 0$, $c_1c_2 > c_3$ for the characteristic polynomial of J; 2×2: trace < 0, determinant > 0). The interval enclosures recorded in the files are degenerate, $[p, p]$. The auditor's verifier, written from scratch, accepted 781/781 files (754 three-species and 27 two-species).

MULTISTATIONARY_CERT. Two distinct exact positive equilibria but no pair with both stable found; 807 orbits. Their witnesses were not audited, so we call them *certified by the producer, not independently verified.* **UNDECIDED:** neither an exclusion nor any witness; 3891 orbits.

5 Results

The undecided fraction 37.0% stayed under the preregistered stop line (50%). The headline `GE2-C-0007` [C] is the table: a *partial* census; only the 754 and 5071 rows are certified, and the 807 are not independently verified.

5.1 Minimal and irreducible networks

`GE2-C-0005` [C]. A bistable network is *minimal* if no proper edge subset is itself BISTABLE_CERT (looked up through canonical forms). In bucket C, 175 of the 727 bistable networks are minimal (the producer's aggregator and a third script by the writer agree; 202 in all buckets). Nine have two

bucket (orbits)	MONOSTAT. (JAC)	BISTABLE_CERT	MULTISTAT._CERT	UNDECIDED
A (681)	245	27	102	307
B (220)	61	0	0	159
C (9622)	4765	727	705	3425
total (10523)	5071	754	807	3891 (37.0%)

Table 1: Census verdicts per bucket (orbits). Labelled counts: JAC 29,948; bistable $162 + 0 + 4338 = 4500$; multistationary-only 4788; undecided 23,033. Sum 62,269. Bucket-A counts equal the two-species control exactly.

internal reactions and are listed in Table 2; the other $718 - 552 = 166$ three-reaction networks of bucket C have no certified bistable proper subnetwork, and 552 of the 718 do.

id	internal reactions	positive equilibrium p	positive equilibrium q
G08823	$2X \to Y$, $Y + Z \to X + Y$	$(983/363, 10/793, 2562/905)$	$(1848/971, 6/955, 3059/923)$
G09468	$2X \to 2Y$, $Y + Z \to 2X$	$(250/869, 18992/653, 628/453)$	$(161/753, 11097/787, 1204/685)$
G09480	$2X \to 2Y$, $Y + Z \to X + Y$	$(95/998, 3385/617, 1337/946)$	$(113/930, 4840/609, 128/101)$
G09927	$2X \to X + Y$, $Y + Z \to 2X$	$(33482/355, 6275/462, 23249/352)$	$(57988/835, 3589/625, 70428/773)$
G09939	$2X \to X + Y$, $Y + Z \to X + Y$	$(89/987, 2465/386, 9690/763)$	$(89/692, 11158/877, 2395/216)$
G10113	$2X \to Y + Z$, $Y + Z \to X + Y$	$(16/841, 150/953, 479/547)$	$(13/972, 3/34, 977/963)$
G10388	$X + Y \to Z$, $X + Z \to 2X$	$(9431/910, 520/803, 25/677)$	$(1567/534, 793/978, 9/206)$
G10467	$X + Y \to 2X$, $X + Z \to 2Y$	$(253/686, 320/313, 15/242)$	$(193/236, 833/858, 11/226)$
G10479	$X + Y \to 2X$, $X + Z \to X + Y$	$(14291/564, 3574/413, 2579/961)$	$(2223/319, 1627/299, 1182/233)$

Table 2: The nine bistable networks with two internal reactions (up to relabelling), with the two stable positive equilibria of one exact witness each. All parameters are positive rationals given in full in `data/witnesses/<id>.json`; every other species flow is as in Section 1. All nine are irreducible in our screen: every species is a reactant somewhere and no equation decouples. This was not checked against the literature.

Cascades versus irreducible mechanisms. Of the 175 minimal bucket-C networks, 79 contain a species that is never a reactant; the other two species then form an autonomous two-species subsystem and the bistability lives in a two-species projection, so these are cascades, not three-species mechanisms. The other 96 are irreducible by this screen (we did not test equivalence with a two-species mechanism with a modified, non-mass-action rate law, which a quasi-steady-state reduction would produce). In bucket C these 79 are also the only bistable networks with a never-reactant species, so none of the 552 non-minimal ones has one. We attach no mechanistic interpretation beyond the listed reactions.

5.2 Inheritance consistency check

`GE2-C-0006` [C]. By Thm 3.1 of Joshi–Shiu [2] ([L]), if a subnetwork N of G has the same stoichiometric subspace and admits multiple exponentially stable positive steady states then so does G; here all flows are present, so the hypothesis holds for every supernetwork. In the census, 609 three-reaction orbits (all buckets) contain a BISTABLE_CERT two-reaction subnetwork: the census certifies 552 of them, has 38 as MULTISTATIONARY_CERT and 19 as UNDECIDED, and none is excluded (no contradiction). By the theorem those 57 are bistable; we do *not* add them to the headline counts, which count only what carries a witness. The check shows that the UNDECIDED and MULTISTATIONARY_CERT classes contain bistable networks, i.e. the 754 is a lower bound for this search, not a count of the bistable networks.

6 Controls

GE2-C-0004 [C]. *Positive control:* the identical pipeline on the two-species class gives exactly 27 BISTABLE_CERT among 681 orbits, the same set (up to S_2) as the bistable networks of Peng's Table 1 [1], and 245 JAC exclusions (Peng: 245; here all ρ are free, so agreement is informative but the criteria differ). Bucket A of the three-species census reproduces 245/27/102/307. *Negative controls:* the empty network and $X \to Y$ and the cycle $X \rightleftarrows Y \rightleftarrows Z$ (four edges, deficiency zero weakly reversible, an external control outside the class) never give BISTABLE_CERT. *Corrupted witnesses:* for an uncorrupted witness the checker passes; with p perturbed by 1/1000, or one ρ multiplied by 1.1, it fails; the auditor's independent verifier does the same and also rejects $+I$ as not Hurwitz. The auditor's numerical falsifier had a positive control (three roots found on witness G01312) and its first-pass "three hits" were duplicates of one root under a too-tight dedupe tolerance.

7 Deviations from the preregistration

Preregistration: data/PREREG_ITEM.md, hash in PREREG_ITEM.md.sha256 (self-attested, no independent timestamp). Three deviations, stated plainly.

1. **N0056 removed as a negative control.** The preregistration listed N0056 ($X \to X{+}Y$, $X{+}Y \to 2X$) as a network that must never be bistable; Peng's Table 1 lists it as bistable, so the preregistered control was wrong.
2. **Stage-1 sampling defect.** The bucket-A control exposed that stage 1 could not sample networks with a species never changed by internal reactions ($p_i = q_i$ is forced, probability zero). The fix (set $q_i = p_i$ for such species) was applied in code and rerun as stage 3 on the 766 affected non-bistable, non-excluded rows: 27 became BISTABLE_CERT and 148 MULTISTATIONARY_CERT. Stage-1 rows remain in the logs; census.jsonl keeps the highest stage (9757 rows from stage 1, 766 from stage 3).
3. **Stage 2 not run; some rules not implemented.** The preregistered stage 2 (10^6 pairs for multistationary-only networks) was not run, and the DIV/ROOT/FLUX rules of Peng were not implemented (they are alphabet-specific).

The three *audit corrections* to the producer's RESULT.md are applied in data/RESULT.md and in this text: (i) bucket C is "connected support 3", not "genuinely three-species"; (ii) the outcome is "754 exact witnesses independently re-verified (781/781 including control); 'new' means new within this class", not checked against Joshi–Shiu-type classifications; (iii) the 807 MULTISTATIONARY_CERT rows are not independently re-verified.

8 Related work and novelty

Read in this work (data/PRIOR_ART_ADDENDUM.md, GE2-L-0002 [L], GE2-X-0002 [X]):

- Peng [1]: two species, at most three internal reactions, inflow and linear removal, 27 of 681 bistable. The two-species classification is Peng's; our two-species run is a control. Peng's methods (nullcline graph, DIV/ROOT/FLUX) are planar-specific.
- Joshi–Shiu [2]: inheritance of stable steady states (Thm 3.1); enumeration of the 386 reversible bimolecular two-non-flow-reaction fully open networks, 35 multistationary, 11 atoms — multistationarity, not stability. Our two-reaction results are a stability statement in the same family; we did *not* compare edge by edge (Section 7 of that paper not read).

- Joshi–Shiu [3]: networks with one species or at most two reactions; Cor. 5.5: at-most-bimolecular networks with exactly two reactions (flows counted) are not multistable. Our networks have six flow reactions, so this does not apply.
- Tang–Xu [4]: smallest multistable networks among two-reaction networks (no flows, no census).

Not read: Siegal-Gaskins et al. (a 2009 census of 40,680 small gene-regulatory networks via network ancestry, cited in [2]; different kinetics), Banaji–Pantea, Banaji–Boros–Hofbauer (read as abstract and grep only by the producer), and the Lixiviant manuscript. **To our knowledge** there is no published census of exact-witness bistability for three-species at-most-bimolecular open networks with at most three internal reactions; this is a recorded search (alphaXiv, web), not a proof of absence, and the authors of the cited papers were not contacted. The method (witness search, determinant-sign exclusion, subnetwork inheritance) is standard.

9 Claim ledger

`ledger/build_ledger.py` files 13 claims in `ledger/ledger.json` with content-addressed evidence in `ledger/evidence/`: [L] (Peng's table; inheritance and small-network results; the determinant-sign tradition), [C] (enumeration; the 754 witnesses; the 5071 exclusions; controls; structure; inheritance consistency; headline `GE2-C-0007`), [X] (the 3891 undecided and the unaudited 807; completeness and novelty not claimed; what was not done). The Elenchus gate (`ledger/run_gate.py`, output in `ledger/gate_output.txt`) reports 13 claims checked, no findings; this is bookkeeping and licenses nothing.

10 Limitations

- **Partial.** 3891 (37.0%) are UNDECIDED. This is search evidence only (sampling 2×10^5 pairs per network); it is never an absence of bistability. The inheritance check shows at least 57 bistable networks lie outside the 754.
- **The** 807 **MULTISTATIONARY_CERT rows are not independently verified.** Sampling audit only for the JAC exclusions ($60 + 40 + 30$ checks); the remaining 5071 rows rest on the producer's run of a proved rule.
- **No claim of new mechanisms.** Some minimal networks are cascades (79 of 175); irreducibility is relative to one screen; no literature check of the nine two-reaction networks beyond Section 8.
- **No biological claim.** The link to genetic switches is motivation only; mass-action kinetics with a shared linear-removal form is a modelling choice, parameters are free positive rationals, and nothing is fitted to data.
- **Novelty is "to our knowledge"** and limited to what was read (Section 8); Section 7 of [2], the cited classifications of [5, 6, 7] and several related preprints were not read.
- **AI producer and one independent AI audit; no human audit**; the preregistration hash is self-attested; no Lean formalisation.
- **Compute.** About 4.5 of the 15 core-hours cap (4.231 for the three-species run in all stages, 0.280 for the two-species control), cores 1–6, detached resumable runs; no reboot detected, wall-clock not tracked precisely.

11 Reproducibility

Repository AutoevolveAI, branch `worktree-openai-math-discovery`; HEAD when this paper was written: commit `1063788`; the paper and data are uncommitted working-tree files at that time. All scripts and results are in `data/` (sha256 in `BUNDLE_MANIFEST.txt`).

```
python3 -I data/code_auditor/verify.py <base-dir>    # exact re-check of every witness (auditor's verifier)
python3 -I data/code_auditor/enum.py ; python3 -I data/code_auditor/jac.py
python3 -I data/code_writer/inherit_check.py data/census.jsonl   # structure + inheritance counts
python3 -I ledger/gen_table.py ; python3 -I ledger/build_ledger.py ; python3 -I ledger/run_gate.py
```

Some scripts take a base directory argument or paths relative to the lane directory (`results/proposal7/2026-11-11/GE2`); the auditor's own run logs were not saved to the bundle.

References

[1] Peng, *Bistability with equal and independent removal in small bimolecular networks*, arXiv:2610.32739 (23 Oct 2026), unrefereed.

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[5] G. Craciun and M. Feinberg, *Multiple equilibria in complex chemical reaction networks: I. The injectivity property*, SIAM J. Appl. Math. 65 (2005) 1526–1546.

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